

Computer Graphics (AR/VR applications), AI & Machine Learning, Mathematics, Deep Learning
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RESEARCH OVERVIEW

My research sits at the intersection of Computer Vision and Agentic AI, focusing on how models perceive, reason about, and act within visual and digital environments. This includes work on inverse graphics, procedural content generation, and computer-use agents that translate visual and expert knowledge into executable actions. I am eager to apply this focus to cutting-edge research in my Master's program.

EDUCATION

Present Sep 2025	University of California, San Diego, California, USA MS in Computer Science Advisor: Manmohan Chandraker Concentration: Computer Vision & Graphics, ML &AI GPA: 3.95/4
May 2025 Nov 2021	Indian Institute of Technology Gandhingar, Gujarat, India B.Tech in Computer Science & Engineering Advisor: Shanmuganathan Raman Major: Computer Science GPA: 3.96/4 dean's list

WORK EXPERIENCE

- Research Assistant, UC San Diego – Centre for Visual Computing, CA, USA (Oct 2025 – Mar 2026)**
 - Conducted research on LLM-based procedural material generation for Blender using expert workflow reasoning. Built automated pipelines to extract structured procedural knowledge from Blender tutorial videos using video understanding and LLM-based analysis.
 - Developed retrieval-time reasoning methods that synthesize expert-inspired material creation processes before compiling them into executable Blender node graphs. Evaluated the system through expert and large-scale user studies, demonstrating improved workflow quality over graph-only procedural generation methods.
 - Publication: Reflecting Process Expertise in Procedural Material Generation: accepted at European Conference on Computer Vision 2026.
 - Advisors: [Manmohan Chandraker](#) (Prof UCSD, [Kunal Gupta](#) (PhD-UCSD)
- ML Intern, Indian Institute of Technology Gandhingar-CVIG Lab, India (Sep 2024-June 2025)**
 - Working on Inverse Modeling for Outdoor Scenes using Single Images
 - Used Gaussian Splatting and Nerf Models to represent Outdoor scenes, and do reconstruction/re-rendering on them.
 - Advisors: [Shanmuganathan Raman](#) (Prof IITGn), [Ashish Tiwari](#) (PhD-IITGn)
- Research Intern, Purdue University -Ideas Lab, West Lafayette, USA (May 2024-July 2024)**
 - Worked on Operator Learning (Neural Operators), for making surrogates for simulation of Fluids (Navier Stokes Equations)
 - Worked on Human Pair motion generation (using Generative Neural Operators)
 - Advisors: [Aniket Bera](#) (Associate Prof. Purdue University, [Hrishikesh Vishwanath](#) (PhD-Purdue University)
- GSoC contributor, Google Summer Of Code [Open Astronomy, Stingray] (May-August 2023)**
 - Developed a Gaussian Processes inference tool that calculates the Evidence of a QPO (Quasi-Periodic Oscillation) Signal in an Astronomical Time Series from violent stellar events like GRBs and magnetar/Solar Flares
 - Made Demonstration and Tutorial Notebooks elaborating on Bayesian Inference and the Project tool.
 - Used Various Python libraries: Jax, TinyGp, Jaxns, tensorflow_probability
 - Open Source Project funded under the Google Summer of Code Program (Gsoc)
 - Pull Request [[webpage](#)] , Final Report [[webpage](#)]
 - Advisors: [Daniela Huppenkothen](#) (University of Amsterdam), [Matteo Bachetti](#) (National Institute for Astrophysics, Italy)

TECHNICAL SKILLS

- **Research Interests:** Probabilistic/Bayesian Machine Learning, Astronomy, Mathematics, Deep Learning, High-Performance Computing, Numerical Analysis
- **Programming Languages:** Python, Jax, C, C++, Julia, Java
- **Other Skills:** Working with Github, Jupyter Notebooks, Google Colab, Unix, Report Writing (Latex, Markdown, Jekyll), HTML

SCHOLASTIC ACHIEVEMENTS

2021-25 **Academic Excellence Award/Dean's list**, Undergraduate in IIT Gandhinagar *India*
2024 Research **Intern Fellowship**, Purdue University, West Lafayette, Indiana *USA*
2021-25 **Awarded the Erach and Meheroo Mehta Merit Scholarship in IIT Gandhingar** *India*
2021 Ranked in **top 0.4 percentile**, IIT/JEE (Advanced) *India*
2019 **Awarded National fellowship**, KVPY, Government of India *India*
2019 **Awarded National Scholarship**, National Talent Search Examination, Government of India *India*

PUBLICATIONS

1. Reflecting Process Expertise in Procedural Material Generation: accepted at European Conference on Computer Vision 2026 [[webpage](#)]
2. Mitigating Simplicity Bias in Neural Networks: A Feature Sieve Modification, Regularization, and Self-Supervised Augmentation Approach (SCSL @ ICLR 2025) [[pdf](#)] [[webpage](#)]
Gaurav Joshi, Parth Narendra Shah, Rachit Verma
3. Stingray 2: A fast and modern Python library for spectral timing (October 2024 JOSS) [[pdf](#)] [[webpage](#)]
Matteo Bachetti, Daniela Huppenkothen, Abigail Stevens, John Swinbank, **Gaurav Joshi**, et al.

KEY PROJECTS

- **PDEs Solving using Physics Informed Neural Networks [Self Project]** *Oct 2023-Dec 2023*
 - Made an implementation of PINNs to solve Laplace Equations for various Dirichlet Conditions
 - Comparative evaluation with analytic and mesh methods.
 - Tools Used: TensorFlow, Latex
 - Final Report [[webpage](#)]
- **Computational Statistics EBook [Jupyter Book]** *July 2023- Oct 2023*
 - Made an Online Book on Computational Statistics which explains the Math behind important Bayesian Data Analysis and Parameter Estimation Methods along with code implementing those methods.
 - Covered important topics like Random Variable Generation, Monte Carlo Integration, MCMC methods (MH, Hamiltonian MC, Gibbs Sampling), Gaussian Processes, Information Theory, Sequential Monte Carlo, Graphical Models, etc.
 - Final Report [[webpage](#)]
- **Effective Sensor Placement using Neural Processes** *July 2023-Aug 2023*
 - Made an ML pipeline that uses Conditional Neural Processes for Effective Pollution Sensor Placement.
 - Used Camx simulated Pollution Data to predict pollution levels for future times and place sensors based on the Mutual Information Criterion.
 - Used and contrasted the quality of fit from various models like Neural Processes, Gaussian Processes, etc.
 - Tools Used: Torch, Jax, DeepSensor
 - Mentor: [Prof. Nipun Batra](#)
- **Scientific Computing EBook [Julia Ebook]** *Mar 2023-May 2023*
 - Made an Online Book on Scientific Computing with Julia
 - Focuses on the Julia Language and Scientific Machine Learning
 - Final Report [[webpage](#)]

KEY COURSES UNDERTAKEN

IIT Gandhinagar, India – Deep Learning, AI & Machine Learning, Probabilistic ML, Natural Language Processing, Data Structures and Algorithms, Advanced Data Structures, Operating Systems, Computer

Vision, Computer Graphics, Data Science, Computer Networks, Numerical Analysis & Computing, Adv Mathematics, Computer Organization and Architecture, Discrete Mathematics.

Purdue University, USA – Algorithms, Computer Graphics, AI & Machine Learning, Computer vision

Additional Information

- Github Profile:-  <https://github.com/Gaurav17Joshi>
- Blog:-  [Blog](#)
- LinkedIn:- [Linkedin Profile](#)

REFERENCES

[Manmohan Chandrakar](#), Professor Visual Computing Center. Contextual Robotics Institutes, UCSD, USA

[Shanmuganathan Raman](#), Jibaben Patel Chair Professor, Computer Science & Engineering, IITGn, INDIA

[Aniket Bera](#), Director of IDEAS Lab, Associate Professor in Computer Science, Purdue University, USA

[Daniela Huppenkothen](#), Staff Scientist, Professor, SRON Institute for Space Research, Netherlands

[Anirban Dasgupta](#), N Rama Rao Chair Professor, Computer Science & Engineering, IITGn, INDIA